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***B.Tech. Degree IV Semester Special Supplementary Examination
September 2014***

CS 1406 DATA COMMUNICATIONS
(2012 Scheme)

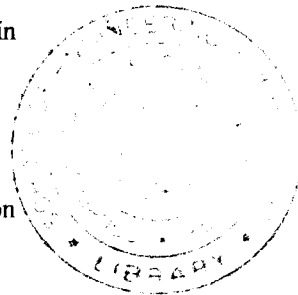
Time : 3 Hours

Maximum Marks : 100

PART A
(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Find the channel capacity of a channel where the spectrum of the channel is in between 3 Mhz and 4 Mhz and SNR = 24dB.
- (b) What are the advantages of digital data transmission over analog data transmission?
- (c) Differentiate between PCM and DM.
- (d) What are the advantages of data compression?
- (e) Why is selective repeat ARQ more suitable in a limited bandwidth communication system?
- (f) What are the transfer modes supported by HDLC?
- (g) Describe the time division multiplexing.
- (h) Explain the concept of spread spectrum with block diagram.



PART B

(4 × 15 = 60)

- II. (a) What are the characteristics of a periodic analog signal? Explain time and frequency domain of composite signal representation. (7)
- (b) Explain various wireless transmission media in detail. (8)
- OR**
- III. (a) Explain various transmission impairments a signal may encounter during transmission. (7)
- (b) Explain nyquist bit rate. Find the nyquist bit rate for a noiseless channel having bandwidth 20KHz and 128 signal levels. (4)
- (c) Explain the use of scrambling with an example. (4)
- IV. (a) Explain various line coding schemes. (7)
- (b) Compare the advantages and disadvantages of parallel and series data transmission. (4)
- (c) Explain a dictionary based compression technique. (4)
- OR**
- V. (a) Explain Huffman coding with a suitable example. (6)
- (b) Explain different digital to analog conversion mechanisms. (9)

(P.T.O.)

- VI. (a) What is CRC? Why does CRC detect more errors than a parity bit? (6)
(b) How is synchronization provided for synchronous data transmission? (5)
(c) Discuss major disadvantages of asynchronous transmission. (4)
- OR**
- VII. (a) Explain various flow control mechanisms available in data communication. (8)
(b) How do link management and flow control achieve in HDLC? (7)
- VIII. (a) Why statistical time division multiplexing is more efficient than a synchronous time division multiplexer? (5)
(b) How is interference avoided using frequency division multiplexing? (5)
(c) Explain CDMA. (5)
- OR**
- IX. (a) What are the advantages of DSL (Digital Subscriber Lines)? Explain various digital subscriber lines. (7)
(b) How are cable networks used in data communication? What is CM and CMTS? (8)
